

Build Embeddings + Similarity Search for Hospitality Documents

Objective:

Build a simple system that converts hospitality-related documents into embeddings and uses a vector database to find documents or information that are most similar to a user's query.

Implementation:

For this task, we created a small set of hospitality documents such as hotel policies, room information, cancellation policies, restaurant details, and guest services. These documents were divided into smaller text chunks so that they could be searched more accurately.

Each text chunk was converted into a numerical representation called an embedding using an embedding model. These embeddings capture the meaning of the text rather than just matching exact keywords.

The generated embeddings were then stored in a vector database. When a user enters a query, the query is also converted into an embedding. The system compares this embedding with the stored document embeddings using similarity search and returns the most relevant results.

Example:

A user might ask:

"What is the hotel's cancellation policy?"

Instead of searching only for the exact words "cancellation policy", the system finds documents that are semantically related, such as:

- Hotel booking and cancellation rules
- Refund policy

- Reservation modification policy

This makes the search more useful because it understands the meaning of the question, not just matching keywords.

Basic Workflow:

Hospitality Documents

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Split into smaller text chunks

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Generate embeddings

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Store embeddings in Vector Database

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User enters a query

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Generate embedding for the query

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Similarity Search

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Return most relevant hospitality information

- **Hospitality Documents** – Collect hotel-related documents such as room details, policies, FAQs, and guest services information.
- **Split into Smaller Text Chunks** – Break the documents into smaller sections so the system can search the information more accurately.
- **Generate Embeddings** – Convert each text section into numbers that represent its meaning.

- **Store Embeddings in Vector Database** – Save these numerical representations in a vector database for quick searching.
- **User Enters a Query** – The guest or staff member asks a question in normal, natural language.
- **Generate Embedding for the Query** – Convert the user's question into the same type of numerical representation.
- **Similarity Search** – Compare the query with the stored embeddings to find the information that is most similar in meaning.
- **Return Most Relevant Hospitality Information** – Show the user the most relevant hotel information found from the documents.

Technologies Used:

- **Python** – Main programming language
- **Embedding Model** – Converts text into numerical vectors
- **Vector Database** – Stores and searches embeddings
- **Similarity Search** – Finds the most relevant documents based on semantic similarity

Outcome:

The system successfully provides relevant hospitality information based on the meaning of the user's query. This approach can later be extended into a hotel AI assistant/RAG system, where the retrieved information can be given to an LLM to generate a natural-language response for guests or hotel staff.